

Cross-validation for time series

DATE

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TOPICS

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R

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I've added a couple of new functions to the [forecast package for R](#) which implement two types of cross-validation for time series.

K-fold cross-validation for autoregression

The first is regular k-fold cross-validation for autoregressive models. Although cross-validation is sometimes not valid for time series models, it does work for autoregressions, which includes many machine learning approaches to time series. The theoretical background is provided in [Bergmeir, Hyndman and Koo \(2015\)](#). So cross-validation can be applied to any model where the predictors are lagged values of the response variable.

This is implemented for NNAR models (neural network autoregressions) in R as follows:

```
modelcv <- CVar(lynx, k=5, lambda=0.15)
print(modelcv)
```

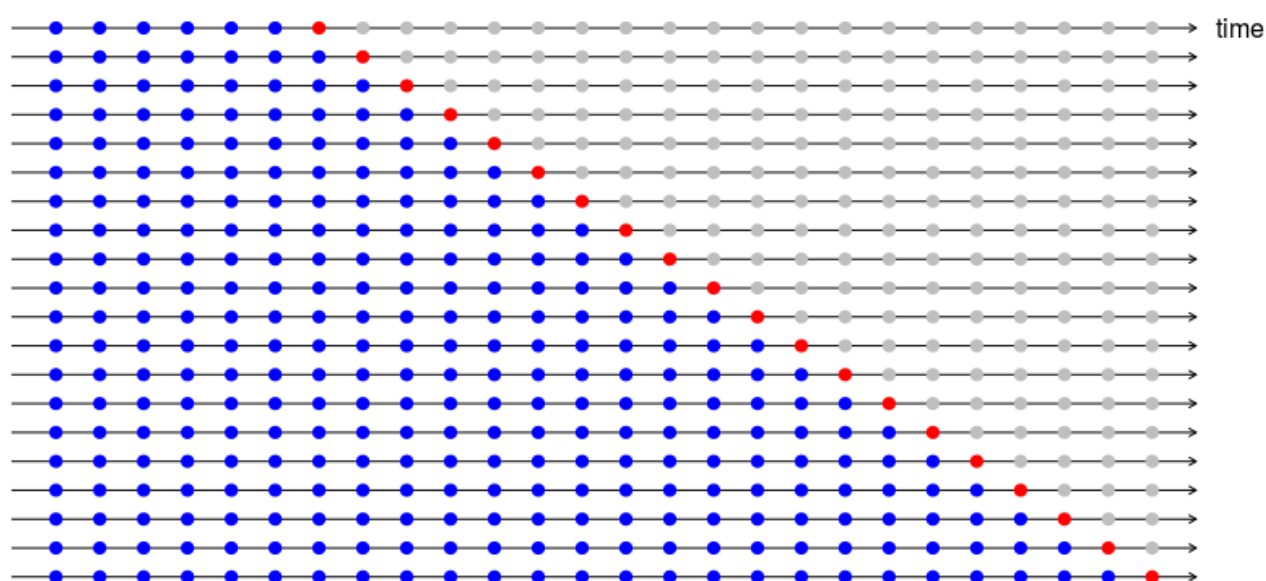
The output is a summary of the accuracy across folds:

```
5-fold cross-validation
              Mean      SD
ME          -32.88142801  98.0725227
RMSE         931.90966858 352.8705338
MAE          608.99488205 272.1244879
MPE          -17.84710226  15.2700638
MAPE          53.99760978  12.7264054
ACF1           0.04842174   0.1480883
Theil's U     0.82984737   0.1487229
```

The `CVar` function is rather limited at this stage, and will only handle cross-validation for models computed using `nnetar`. If there is enough interest, I might add more functionality at a later stage.

Time series cross-validation

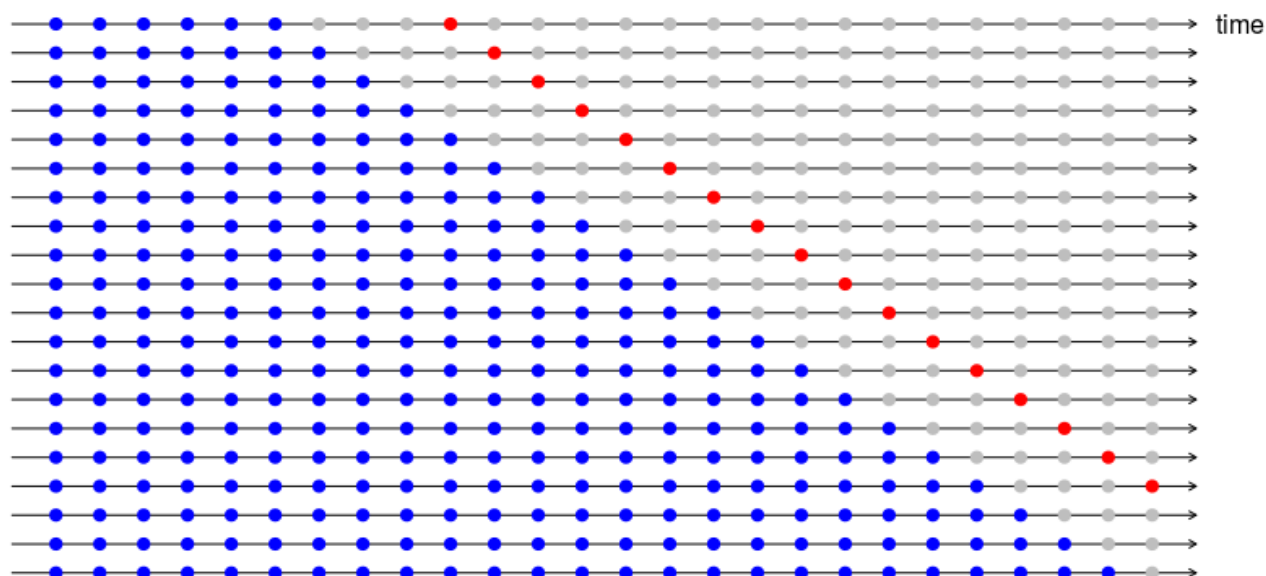
In this procedure, there is a series of test sets, each consisting of a single observation. The corresponding training set consists only of observations that occurred *prior* to the observation that forms the test set. Thus, no future observations can be used in constructing the forecast. The following diagram illustrates the series of training and test sets, where the blue observations form the training sets, and the red observations form the test sets.



cv1-1

The forecast accuracy is computed by averaging over the test sets. This procedure is sometimes known as “evaluation on a rolling forecasting origin” because the “origin” at which the forecast is based rolls forward in time.

With time series forecasting, one-step forecasts may not be as relevant as multi-step forecasts. In this case, the cross-validation procedure based on a rolling forecasting origin can be modified to allow multi-step errors to be used. Suppose that we are interested in models that produce good 4-step-ahead forecasts. Then the corresponding diagram is shown below.



cv4-1

Time series cross validation is implemented with the `tsCV` function. In the following example, we compare the residual RMSE with the RMSE obtained via time series cross-validation.

```
library(fpp)
e <- tsCV(dj, rwf, drift=TRUE, h=1)
sqrt(mean(e^2, na.rm=TRUE))
## [1] 22.68249
sqrt(mean(residuals(rwf(dj, drift=TRUE))^2, na.rm=TRUE))
## [1] 22.49681
```

Here I apply a random walk with drift to the Dow-Jones index time series `dj`. The first calculation implements a one-step time series cross-validation where the drift parameter is re-estimated at every forecast origin. The second calculation estimates the drift parameter once for the whole data set, and then computes the RMSE from the one-step forecasts. As expected, the RMSE from the residuals is smaller, as the corresponding “forecasts” are based on a model fitted to the entire data set, rather than being true forecasts.

The `tsCV` function is very general, and will work for any forecasting function that returns an object of class `forecast`. You don’t even have to specify the minimum sample size for model fitting, as it will silently fit models beginning with a single observation, and return a missing value whenever the model cannot be estimated.

These functions are only on the [github version](#) for now, but will migrate to [CRAN](#) next time I release a new stable version.

0 reactions



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Oldest

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rrremedio [Jan 6, 2023](#)

"Finally, we compare the forecasts against the actual values and average over the folds."
and "Finally, we can compare the forecasts against the actual values, averaged across each forecast horizon."

When using the "by" argument in "accuracy" the accuracy measures are averaged?

↑ 1



1 reply



robjhyndman [Jan 6, 2023](#)

"by" is essentially grouping the data by whatever variables are specified, before averaging across the accuracy measure within each group.



aerazov [Apr 25, 2023](#)

I have a time series of 1000 daily observations, and I want to do an evaluation on a rolling forecasting origin, how many series of cross validation should I do for an $h=10$?

↑ 1



2 replies



robjhyndman [Apr 25, 2023](#)

The forecast horizon isn't really the issue. You should do as many folds as you can afford to given the computational cost.



tsbreed [Jun 7, 2023](#)

Are there any research on this that you are aware of? I mean, that as many folds as possible is preferable?

For example, say I want to forecast CPI inflation for two years ahead of which I have monthly data from 2000-2023. I want to use tsCV to compare some models but where (when) is it optimal to set my first origin? There doesn't seem to be much value in starting my tsCV from June 2000, as the training set would then only consist of five months. So where do I set my first origin? Furthermore, do I really want to let the forecasts from, say, 2005-2010, to affect my evaluation of a model that is supposed to forecast from 2023 to 2025? Wouldn't it just be better to use all years before 2018 or so as a training set, and maybe use January 2018 as my first tsCV origin?



NikTuzov [Jul 4, 2023](#)

Dear Prof. Hyndman:

Thank you for a great tutorial.

ARIMA and ETS are adaptive, they search for the best model, meaning that the fitted model depends on the data set.

E.g. for the beer case, there are 207 training data sets and there is no guarantee that the same model is used on each.

Do you think it's better to specify the model fully to make sure it's always the same?

Regards,

Nik

↑ 1



3 replies



robjhyndman [Jul 4, 2023](#)

It depends on what you are testing. By allowing the model to change with the training set, you are evaluating the performance of the algorithm, including the model selection. If you restrict the model to be the same for all training sets, then you are evaluating the performance of that specific model. It is not obvious how to choose

one model in this context. If you choose the model that is selected using all the data, then you are using future information to some of the training sets, and then the forecasts are not really forecasts. If you choose the model that is selected for the smallest training set, then it may not be a good model for the larger training sets.



NikTuzov [Jul 5, 2023](#)

Thank you for replying.

About evaluating the performance. For instance, if we use MAE, then, since different models are fitted on different folds, the expected value of AE is also dependent on the fold. Therefore, when we compute the final MAE, we average over a number of realizations of random variables with possibly distinct expected values.

Moreover, even if the model is not adaptive, that issue is still present. Suppose the true model is ARMA(1, 1) and we specified it as such. As the number of observations in the training data set goes up, the estimated parameters converge to the true values. Hence, the expected AE is going to be lower for the folds that have more observations.

Could you possibly refer me to some books or papers where this issue is considered in detail.

Regards,
Nik



robjhyndman [Jul 5, 2023](#)

The error due to parameter estimation is much smaller than other sources of error, so provided the initial folds are not too small, we can ignore the effect of estimation error. I don't know any good references on this.



NikTuzov [Jul 13, 2023](#)

Apart from TSLM(), are there any models in fable that take the uncertainty of estimating the regression coefficients into account when computing the forecast variance?

↑ 1 

1 reply



robjhyndman [Jul 13, 2023](#)

Forecasts from NAIVE and RW also allow for uncertainty in the drift coefficient.

  1



MRibeiro2107 [Aug 23, 2023](#)

When I using the code:

```
#TSCV datasets
```

```
dados.ts.tr <- dados.ts %>%
```

```
stretch_tsibble(.init = 3, .step = 1)
```

```
dados.ts.tr
```

```
#Accuracy
```

```
temperatura.fit.tscv<- dados.ts.tr %>%
```

```
model( Seasonal naïve TSCV = SNAIVE(Temperatura)) %>%
```

```
forecast(h = 1)
```

```
View(temperatura.fit.tscv)
```

```
acc.sn.tscv<- temperatura.fit.tscv %>%
```

```
fabletools::accuracy(dados.ts)%>%
```

```
mutate(Method = paste(.model, "method")) %>%
```

```
select(Method, RMSE, MAE, MAPE)
```

In the results the first 9 rows of temperatura.fit.tscv objet in .mean column are missing. Is there any reason for that?

↑ 1 

2 replies



robjhyndman [Aug 24, 2023](#)

I'm guessing your data is monthly, so a seasonal naive method needs 12 months of data to produce a forecast. With `.init=3`, that means the first 12-3 rows correspond to situations where a model cannot be produced. You need to set

`.init=12` with this method and this type of data.



MRibeiro2107 [Aug 24, 2023](#)

Thank you, Rob.

Yes, I have monthly data.



marczeman [Aug 28, 2023](#)

I fit multiple series (hierarchical), and get back a mable with one row per series:

```
lung_deaths_agg <- as_tsibble(cbind(mdeaths, fdeaths)) %>%  
  aggregate_key(key, value = sum(value))
```

```
lung_deaths_agg_train <- lung_deaths_agg |> filter(year(index) < 1979)
```

```
fit <- lung_deaths_agg_train %>%  
  model(auto_lm = TSLM(value ~ trend() + season())  
        ,auto_arima = ARIMA(value))
```

How do I replace one particular model, if I don't like the automatically fit one? let's say, in particular I want to replace the `fdeaths` arima fit to my own custom specified one, before I pass this mable to be reconciled?

↑ 1



2 replies



robjhyndman [Aug 29, 2023](#)

Here is an example:

```
fit2 <- lung_deaths_agg_train |>  
  filter(key == "fdeaths") |>  
  model(myarima = ARIMA(value ~ pdq(2,0,0)))  
fit <- fit |>  
  left_join(fit2) |>  
  mutate(auto_arima = if else(key == "fdeaths", myarima, auto_arima))
```



```
select(-myarima)
```



marczeman [Aug 29, 2023](#)

Thanks! That worked perfectly.



RishavDaredevil [Sep 18, 2023](#)

Dear Prof. Hyndman:

Thank you for this. I had a question regarding the `as_fable` function. What does **response** refer to here as with the dataset as I was working on I thought it would be the point forecast which means I would have inputted `response = '.mean'`. But as I saw in the book and here in the examples it always refers to the **distribution column name in quotation**

```
# A fable: 900 x 6 [1Q]
# Key:      .id, .model [225]
  .id .model Quarter      Trips .mean      h
  <int> <chr>   <qtr>      <dist> <dbl> <int>
1     1 rw     1999 Q3  N(736, 38752)  736.     1
2     1 rw     1999 Q4  N(721, 90421)  721.     2
3     1 rw     2000 Q1  N(705, 155007)  705.     3
4     1 rw     2000 Q2  N(690, 232510)  690.     4
5     1 naive  1999 Q3  N(751, 26066)  751.     5
6     1 naive  1999 Q4  N(751, 52131)  751.     6
7     1 naive  2000 Q1  N(751, 78197)  751.     7
8     1 naive  2000 Q2  N(751, 1e+05)  751.     8
9     1 snaive 1999 Q3   N(954, 6018)  954.     9
10    1 snaive 1999 Q4   N(935, 6018)  935.    10
#  890 more rows
#  Use `print(n = ...)` to see more rows
```

Regards,
Rishav



1 reply



robjhyndman [Sep 18, 2023](#)

The response variable is the variable you are modelling. It is usually the same as the name of the distribution column in the forecast object.



mariospelekanos [Oct 4, 2023](#)

Dear Professor,

According to the definition of tsCV function, if exogenous predictors are used then xreg and newxreg arguments correspond to the training and test periods. How will it be implemented with a leave k-out cross validation (monthly data) for a multi-step forecast? xreg will be defined for the same time period as the time series and newxreg for the forecasting period ahead?

Thank you in advance!

↑ 1



1 reply



robjhyndman [Oct 4, 2023](#)

Yes, the xreg values for the corresponding time periods are used.



kaituna [Mar 5, 2024](#)

Hi there,

Is there an easy way to fit multiple models on a hierarchical dataset (in my case the key columns are 'provider' and 'test') and then select the best model for each key combination and produce forecasts for only those models? I'm trying to use the accuracy measures to identify the best model in each case then create a new column in the mable called 'best_model' but it doesn't seem to be working.

↑ 1



5 replies



robjhyndman [Mar 5, 2024](#)

Here's an example: <https://community.rstudio.com/t/extracting-best-models-from-a-mable/54243/4?u=robjhyndman>



kaituna [Mar 6, 2024](#)

edited

Thanks for the reply and the link.

I think I already went down this path but I have 5 models, rather than just two so the 'ifelse' method didn't seem usable.

Instead, I identified the best model for each series and then joined that table to the mable so I have 'best_model' column:

```
fc_accuracy <- fabletools::accuracy(fc, data_hts,
                                   measures = list(
                                     point_accuracy_measures,
                                     interval_accuracy_measures,
                                     distribution_accuracy_measures
                                   )
                                   )

model_performance <- fc_accuracy %>%
  filter(is_aggregated(provider)) %>%
  group_by(modality, .model) %>%
  summarise(
    RMSE = mean(RMSE),
    MAE = mean(MAE),
    MAPE = mean(MAPE),
    Winkler = mean(winkler),
    CRPS = mean(CRPS)
  ) %>%
  arrange(MAPE)

#build table of best models based on MAE
best_models <- model_performance %>%
  group_by(modality) %>%
  filter(MAE==min(MAE)) %>%
  select(modality, .model)

best_fit <- fit %>%
```

```

filter(is_aggregated(provider)) %>%
left_join(best_models, by = join_by(modality))

names(best_fit)[9] = "best_model"

```

My intention was to then just copy the value from the relevant model column on any given row into a new column called 'selected_model' based on what was in the 'best_model' column (i.e. if the 'best_model' was snaive then I'd copy the contents of that row's 'snaive' column)

```
best_fit <- best_fit %>% mutate(selected_model = purrr::map2(row_number(
```

This doesn't work though as I don't seem to get a valid model inserted into the 'selected_model' column. Instead of it being a 'lst_md1' type it is a 'List':

```

str(best_fit, max.level=2)
mdl_df [16 x 10] (S3: mdl_df/tbl_df/tbl/data.frame)
 $ provider      : chr* [1:16] <aggregated>, <aggregated>, <aggregated>,
 $ modality      : chr* [1:16] AUDIOLOGY_ASSESSMENTS , BARIUM_ENEMA
 $ base          : lst_md1 [1:16]
 $ snaive        : lst_md1 [1:16]
 $ ets           : lst_md1 [1:16]
 $ arima         : lst_md1 [1:16]
 $ mixed         : lst_md1 [1:16]
 $ ols           :List of 16
  .. attr(*, "method")= chr "ols"
  .. attr(*, "sparse")= logi TRUE
  .. attr(*, "class")= chr [1:3] "lst_mint_md1" "lst_md1" "list"
 $ best_model     : chr [1:16] "snaive" "ols" "mixed" "arima" ...
 $ selected_model:List of 16
 - attr(*, "response")= chr "value"
 - attr(*, "key")= tibble [16 x 3] (S3: tbl_df/tbl/data.frame)
  .. attr(*, ".drop")= logi TRUE
 - attr(*, "model")= chr [1:6] "base" "snaive" "ets" "arima" ...

```

Any ideas on how I can copy the identified model from the original column into the 'selected_model' column?



robjhyndman [Mar 7, 2024](#)

You can use a `case_when` statement, as shown in this example:

<https://gist.github.com/robjhyndman/86c8a26bcde0313ad02b9856c7b5bad1>



kaituna [Mar 8, 2024](#)

Thanks, that worked for me. I'm curious why my method resulted in a list rather than `lst_md` data type but I guess that is an R issue rather than a fable specific one. It's a pity as I was looking for a more general solution that wouldn't require the underlying code to be changed if I used different models, but for now I'm getting what I need.

Is there anything on the fable development roadmap to automate the selection of models from a range of possibilities?



robjhyndman [Mar 8, 2024](#)

I've updated the gist to be more general, along the lines you were attempting.

Yes, we have thought of adding tools to make this easier for users. However, it's not high on the list compared to the current work being done around rewriting the reconciliation parts of fabletools.



ustcdj [Mar 15, 2024](#)

For grouped time series (gts), it looks like the model (pdq)(PDQ)m and exogenous features coefficients from gts modeling with ARIMA are the same as using regular ARIMA. But the point estimate forecast results are different since gts will reconcile the ARIMA forecast while regular ARIMA will not. How to find information about the reconciliation? Or how to use reconciliation information, combined with the point forecast from ARIMA to calculate the final reconciled point forecast? I know the gts (with ARIMA) forecast can give the point forecast results directly. How to get them manually? Thanks!

↑ 1



4 replies



robjhyndman [Mar 16, 2024](#)

It sounds like you already know how to compute ARIMA forecasts, and how to compute reconciled forecasts using ARIMA models. If you want to do the latter in two steps (i.e, compute the ARIMA forecasts first, and then reconcile them), you could use the `hts::combinef()` function.



dj **ustcdj** [Mar 17, 2024](#)

Thanks for the quick response! I haven't been able to install hts pkg due to error msg like "cannot compile rcpparmadillo" etc. Before I figure it out, just want to confirm that `hts::combinef()` will show me the reconciliation parameters that I can use to calculate the final reconciled forecast? I know how to get ARIMA forecasts using `(pdq)(PDQ)m` and coefficients, such as below. What I need are the parameters for the reconciliation process.

BTW: would it be a good idea to have `combinef()` in fable?
Thanks again!

sample auto ARIMA output below

```
Series: log(mydata_ts)
ARIMA(2,1,1)(1,0,0)[12]

Coefficients:
          ar1          ar2          ma1          sar1
      -1.1413   -0.3872    0.9453    0.7572
s.e.    0.1432    0.1362    0.0593    0.0830

sigma^2 estimated as 0.006575:  log likelihood=48.35
AIC=-86.69  AICc=-85.23  BIC=-77.44
```



robjhyndman [Mar 17, 2024](#)

I don't know what you mean by the "reconciliation parameters". Reconciliation is done via a matrix G (see <https://otexts.com/fpp2/reconciliation.html>). That matrix is not parameterised explicitly, but depends on how the covariance matrix of the one-step base forecast errors is estimated. It is not returned by `hts::combinef()` as it is usually much too large, and of limited interest in itself. But if you want to compute it

usually much too large, and of limited interest in itself. But if you want to compute it, you can use the equations in the link above.



dj **ustcdj** [Mar 17, 2024](#)

The matrix G is exactly what I am after. Appreciate your help!



dj **ustcdj** [Mar 31, 2024](#)

When there are external regressors, I found it hard to manually find $\text{ndiffs}/\text{nsdiffs}$ (d/D) for ARIMA. The d/D values may change depending on how many and what combinations of external regressors are included. How does fable come up with $\text{ndiffs}/\text{nsdiffs}$ (d/D) for ARIMA when there are external regressors? According to the book [Regression with ARIMA errors using fable](#),

The `ARIMA()` function can also be used to select the best ARIMA model for the errors. This is done by not specifying the `pdq()` special. Whether differencing is required is determined by applying a KPSS test to the residuals from the regression model estimated using ordinary least squares.

Given `ARIMA()` can automatically come up with parameters, I feel it's much easier for `ARIMA()` to figure out the d/D values when using external regressors. And the traditional way to estimate the d/D with visuals or tests on dependent variable will not work. Am I correct?

If differencing is required, then all variables are differenced and the model re-estimated using maximum likelihood estimation.

Sometimes, the level of differencing on dependent variable and external regressors may not match. How does fable handle that?

Thanks!

↑ 1



1 reply



robjhyndman [Apr 1, 2024](#)

There is no need to manually apply `ndiffs` or `nsdiffs` to any series. As per the

quote you give, the number of differences used by `ARIMA()` is based on a KPSS test of the **residuals** of the model. That is, it fits a regression model, and applies `ndiffs()` to the residuals to determine the number of differences to apply to all series (response series and all covariates). It is possible for all series to be nonstationary, but the residuals to be stationary -- this is an example of cointegration. In such cases, you don't need to difference the series before fitting the model. This also shows why just determining the differencing by looking only the response series is not a good idea, as it will over-difference cointegrated series.



dj **ustcdj** [Apr 2, 2024](#)

I am running a model `stepwise=FALSE, APPROX=FALSE` with over 100 observations. Somehow, I got a message, "Warning: NaNs produced". For terms, such as `sar2` or `sma1`, they have estimate, but NaN for `std.error`, `statistic`, and `p.value`. According to a similar question on [stackoverflow](#), lower `max.order` is suggested. What is the root cause of this? Can I use the estimate, even with the warning message? Thanks, again!



1 reply

 **robjhyndman** [Apr 2, 2024](#)

If there are NaNs produced, the model is potentially numerically unstable and should probably be avoided. You could try to use the model and see whether the forecasts are any good. Without reproducible code, it is impossible to say why this has occurred for a specific series.



dj **ustcdj** [Apr 4, 2024](#)

In fable, how to do backtesting on more than one-step, say $h=12$ months of training data after a model is trained on the full ten years of training data? According to this thread, [Does the forecast function within fable provide one-step forecasts?](#), it seems one-step accuracy is provided. How to get $h=12$ step accuracy? Thanks!

The `fitted()` values of a model are one-step ahead forecasts, which can be used to

evaluate 'training accuracy' performance (forecast accuracy on the training data). However there's a catch - the estimated parameters of the model are based on the entire training set and so the training accuracy is better than what can be expected (the model contains some information about the future it is fitting).

↑ 1 

1 reply



robjhyndman Apr 4, 2024

There is an example at <https://otexts.com/fpp3/tscv.html#example-forecast-horizon-accuracy-with-cross-validation>

  1



RodolfoViegas Feb 6

Prof. Hyndman, my question is about using machine learning models and cross-validation. Well, for time series, cross-validation is used to evaluate models' performance, in machine learning too, but also to do hyperparameter tuning. If I do the classical splitting into two sets, training set and test set, and then tuning my model -- let's say a MLP -- with grid search. Inside the hyperparameter searching/testing there is a cross-val method. Ok, done the proceeding, could (or should) I do cross-validation on test set without refitting the model? I question this because your example the model needs to retrain window after window for make forecasts, and I do not know if I did it, without retraining, would invalidate the process. And also I don't know if I really should do cross-validation on test set

↑ 1 

2 replies



robjhyndman Feb 6

In the types of models I use, we don't need to use a different set of data for hyperparameter tuning. Instead, we can use the AIC or AICc. That is why we don't discuss validation sets in <https://Otexts.com/fpp3>. But if you were using a method that did need a validation set for hyperparameter tuning, you could use a nested cross-validation. e.g., <https://arxiv.org/abs/1809.09446>.

If you used time series cross validation without retraining the model in each window, you would not get invalid results, but you would only be measuring the accuracy of

that particular model, not the accuracy of a model fitted to the data immediately before the test set.



RodolfoViegas [Feb 8](#)

Thanks, Professor.



Charles-cod22 [Aug 23](#)

Dear Professor,

I have some challenge with evaluating the performance of the model `auto.arima()` or `Arima()` gives in the aspect of training, testing for prediction in relation to a univariate time series analysis. For instance, the data I am analysing is looking like this:

We Perform the model evaluation

We divide the data set into train and test data

And then fit both data and apply prediction on both

```
traindat = window(x1, end = 25)
testdat = window(x1, start = 26)
fit = auto.arima(traindat)
summary(fit)

fr_test = forecast(fit, h = length(testdat))
plot(fr_test)
points(testdat)
accuracy(fr_test, testdat)
```

```
accuracy(fr_test, testdat)
```

```
plot(fr_test)  
lines(testdat)
```

Are these steps right or wrong?

Thank you for your help.

Best

↑ 1



3 replies



Charles-cod22 [Aug 23](#)

And can do the model performance evaluation using tsCV and how?



robjhyndman [Aug 23](#)

Yes, you can evaluate a model on a test set like that. But it is better to use time series cross-validation as it evaluates over a much larger set of observations. Here is an example computing MSE by horizon:

```
e <- tsCV(x1, forecastfunction = function(y, h) {  
  fit <- auto.arima(y)  
  forecast(fit, h = h)  
}, h = 5)  
  
colMeans(e^2, na.rm = TRUE)
```



Charles-cod22 [Aug 24](#)

Alright Professor, thank you.



AlexeiUjegov [Sep 1](#)

Dear Prof. Hyndman,

thank you for great materials. What is the best way to choose a set of exogenous regressors for scenario forecasting?

E.g. I need to estimate ARX model. Let it be $K = 20$ exog regressors is given. I need to choose a set of $j = 5$ exog regressors that give best accuracy during cross-validation. Exog regressors can be lagged exog variables and lagged dependant variable.

Solution that comes to mind to iterate all possible sets of 5 regressors to choose the best according to CV.

I do understand how to perform CV for exact model specification, but have no idea how to iterate model specification using fpp3 (tidyverts, forecast packages).

A simple example or tips would be great

Thank you!

↑ 1



1 reply



robjhyndman [Sep 1](#)

We don't have any easy user-facing functions to do that. Here is the code used to produce Table 7.1 in fpp3 (<https://otexts.com/fpp3/selecting-predictors.html#tab:tblusMR>):

```
library(fpp3)
library(rlang)
library(purrr)

opts <- expand.grid(Income = 0:1, Production = 0:1, Savings = 0:1, Unemp
mutate(
  formulae =
    pmap(
      list(Income = Income, Production = Production, Savings = Savings
function(...) {
  spec <- list(...)
  new_formula(
    sym("Consumption"),
    reduce(syms(names(spec)[spec == 1]), call2, .fn = "+", .init
  )
}
```

```

    ),
    models = set_names(map(formulae, TSLM), map_chr(formulae, deparse))
  )

tab <- us_change |>
  model(
    !!!opts$models
  ) |>
  glance() |>
  bind_cols(opts) |>
  transmute(Income, Production, Savings, Unemployment, AdjR2 = adj_r_squ
  arrange(AICc)

tab

```

You could modify that for your own situation.



Charles-cod22 [Sep 8](#)

Dear Professor,

How do I assess the VAR(2) model performance with train and test data in R? For example,
`D = cbind(x5, x6)`

```

m56_var2 = VAR(D, p = 2, type = "both")
summary(m56_var2)

```

```

Var_forcas = predict(m56_var2, n.ahead = 5)
head(Var_forcas)

```

```

train.dat = window(D, end = 21)
test.dat = window(D, start = 22)

```

```

fit.Var = VAR(train.dat, p = 2, type = "both")
summary(fit.Var)

```

```

x56_pred = forecast(fit.Var, h = 9)

```

```

plot(x56_pred, xlab = "Time (years)")
points(rep(c(22:30), 2), test.dat)

```

accuracy(x56_pred, test.dat, D = NULL, d = NULL) Does these give the right way for VAR model evaluation?

Thank you Prof.

↑ 1



4 replies



robjhyndman [Sep 8](#)

This is a post on the fable package. You appear to be using the forecast package. Otherwise, it looks ok.



Charles-cod22 [Sep 8](#)

Okay. Thank you Prof.

But in place of function "forecast" what can I use? Is it the predict function or some new function?



robjhyndman [Sep 8](#)

You can use the forecast function from the forecast package, which is exactly what you are doing.



Charles-cod22 [Sep 9](#)

Thank you Prof.



Charles-cod22 [Sep 9](#)

Dear Prof,

I have some challenge here.

I have used lm() to fit AR(2) + trend model in R. Now I want to make predictions but it

messing up the whole thing.

You may have to help me here. Here are my sample code:

Analysis for data x3:

```
y3d = x3[-c(1,2)] # start
y3l1 = x3[-c(1,30)] # l1
y3l2 = x3[-c(29,30)] # l2
cons = rep(1, 28) # const term
tre = 1:28 # trend term
Y33 = cbind(y3d, y3l1, y3l2, tre)
conames = c("yy", "yyl1", "yyl2", "trd")
y.data = data.frame(Y33)
colnames(y.data) = conames
head(y.data)

f.lm2 = lm(yy ~ ., data = y.data)
summary(f.lm2)
```

Predict with data points

```
x.val2 = round(seq(max(x3), min(x3), length = 28))

y.val2 = predict(f.lm2, newdata = data.frame(x3 = x.vals2))

Pred.O_Tb = ts(y.vals2, start = c(1996, 1), frequency = 1)
y.b2 = cbind(Pred.O_Tb, y3d)

ts.plot(y.b2, xlab = "Time (years)", ylab = "Cases of P-Tb",
main = "AR(2) Predictions of x2 Data Points", col = c(5,7))
legend("topright", legend = c("O-Tb", "Pred O-Tb"),
lty = 1, lwd = 2, col = c(5,7))
```

The error I get is this: Error in eval(predvars, data, env) : object 'yyl1' not found. I have tried to modify my codes but the issue remains.

What is your advice for correct this?

Thank you.

↑ 1



1 reply



Charles-cod22 [Sep 9](#)

Professor thank you I just fixed the issue.



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Aa

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